# xphd

## Exciton–phonon linewidths, symmetry and real-time dynamics from Quantum ESPRESSO, LetzElPhC and yambo

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**What this document is.** A guide to the `xphd` package and to the calculations built on it, for planar monolayer GaN and monolayer hBN. We give every command in the order it is run, what it needs, what it writes and how to tell whether its output is right. We also describe the upstream Quantum ESPRESSO, LetzElPhC and yambo settings the code relies on, the conventions it uses, the problems I met along the way, and finally the physics and numerics behind each result. The numbers we quote were checked against our own outputs, so you can use them to test your installation.

## Contents

**1 What xphd does, and how to install it** 1
1.1 Install . . . 2

**2 The workflow at a glance** 2

**3 Conventions the code relies on** 3
3.1 Units and indices . . . 3
3.2 Electron–hole pairing . . . 3
3.3 Mirror parity . . . 3

**4 Data files** 3

**5 Command reference** 4
5.1 Scripts . . . 5

**6 Upstream settings** 6
6.1 Quantum ESPRESSO: ground state and phonons . . . 6
6.2 q2r.x and matdyn.x on the $360 \times 360$ mesh . . . 6
6.3 LetzElPhC . . . 7
6.4 yambo: GW and BSE . . . 7
6.5 From databases to archives . . . 7

**7 Worked example: planar monolayer GaN** 8
7.1 Commands, in order . . . 8

**8 Worked example: monolayer hBN, the bright-minimum contrast** 11
8.1 Commands, in order . . . 11

**9 Pitfalls we found, and what fixed them** 13

**10 Methods and derivations** 13
10.1 Notation . . . 13
10.2 Exciton–phonon matrix elements . . . 14
10.3 Mirror parity and selection rules . . . 14
10.4 Near-degenerate excitonic spaces . . . 16
10.5 Exciton–phonon linewidths . . . 17
10.6 Phonon-assisted luminescence and helicity . . . 18
10.7 Real-time exciton Boltzmann transport . . . 19
10.8 Validation summary . . . 20

**References** 21

## 1. What xphd does, and how to install it

`xphd` (exciton–phonon dynamics) takes the output of a Quantum ESPRESSO, LetzElPhC and yambo calculation and turns it into five things. First, exciton–phonon matrix elements on the full Brillouin zone, stored as one archive for every

exciton momentum $\mathbf{Q}$. Second, linewidths computed without any artificial broadening. Third, the mirror ($\sigma_h$) parity of electrons, phonons and excitons, together with the selection rules it implies. Fourth, exciton dipoles, helicity and phonon-assisted luminescence. Fifth, a real-time Boltzmann transport equation (rt-BTE) for the exciton populations. A set of diagnostics tells you whether each of these numbers can be trusted.

We wrote it for planar two-dimensional semiconductors, and we developed and tested it on monolayer GaN and monolayer hBN. This guide explains how to run it, what every step needs and writes, and how to check each result; Sec. 10 then explains the physics and numerics behind each quantity, so that you can judge what the code is doing rather than take it on trust.

### 1.1 Install

```
cd xphd
pip install -e ".[dev,plot]"
xphd --help                 # lists every subcommand
xphd <command> --help       # every command documents its own options
python -m pytest            # runs on synthetic data, no yambo databases needed
```

Only `numpy` is a hard requirement. `scipy` is needed for the transport equation; `yambopy`, `netCDF4` and `tqdm` are needed by anything that reads yambo or LetzElPhC databases; `matplotlib` is used for figures. All of these are imported only when a command needs them, so `-help` always works.

There are two kinds of tools. *Subcommands* (`xphd <name> ...`) run from the command line. *Scripts* (in `scripts/`) are copied into your working folder, edited in their `SETTINGS` block at the top and run as ordinary Python files; we chose this style so that each analysis is a short, readable file you can keep next to your data.

> **Line continuation.** Long commands are wrapped here with a trailing \ (bash). In PowerShell, either type the command on one line or end each wrapped line with a backtick. Paths in angle brackets, such as `<SAVE>`, are placeholders for your own folders.

## 2. The workflow at a glance

Every material goes through the same stages. Stages A–C are the upstream codes; `xphd` starts at stage D. Once the archives exist, the analysis branches E–I are independent of one another.

| Stage | Code | Input → output | Notes |
|---|---|---|---|
| **A. Ground state** | `pw.x` scf, then nscf; `p2y` | structure → `<prefix>.save`, yambo `SAVE` | nscf on the full $N \times N$ k-mesh with enough bands for the screening. Keep scf and nscf in separate directories, because the nscf overwrites `<prefix>.save`. |
| **B. Phonons** | `ph.x`, `q2r.x`, `matdyn.x` | scf save → dyn files, `dvscf`, force constants, fine-mesh frequencies | q-grid equals k-grid ($N \times N$). `loto_2d=.true.` and the crystal acoustic sum rule in `q2r` and `matdyn`. Fine mesh $360 \times 360$ through the chunked `run_matdyn.sh`. |
| **C. Excitons** | yambo (GW optional) | `SAVE` → `ndb.BS_diago_Q1...` | Bethe–Salpeter equation at every irreducible $\mathbf{Q}$, full diagonalisation. |
| **C′. Electron–phonon** | LetzElPhC | `SAVE` + `ph.save` → `ndb.elph` | Must use the same `SAVE` as the BSE, so that both share one wavefunction gauge. |
| **D. Archives** | `xphd dmats`, `generate`, `check-archive`, `verify-archives` | → `Dmats.npy`, `GI_ExcPh_Q0001.npz...` | Build the $\Gamma$ archive first; `check-archive` must print READY before you build the rest. |
| **E. Linewidths** | `xphd matdyn`, `linewidth`, `sweep`, `unfold` | → `hw_fine.npy`, text tables, zone fields | Broadening-free triangle integration on $360 \times 360$. |
| **F. Symmetry** | `xphd parity`, `band-parity`, `irreps`, `selection-rule`; `labels_from_elph.py` | → `parity.npz`, `mode_labels_elph.npy` | Parity of every state everywhere; the zeros are set by symmetry. |
| **G. Optics** | `xphd dipoles`, `chirality` | → `exc_dipoles.npy` | Complex $x, y$ dipoles from one BSE. |
| **H. Dynamics** | `xphd bte`, `bte-check`; `check_bte.py`, `valley_times.py` | → snapshots | Needs every archive of the full zone. |

# 3. Conventions the code relies on

## 3.1 Units and indices

- Energies are in eV inside archives and `npz` files. Printed tables use meV where they say so. The `matdyn` frequencies (cm$^{-1}$) are converted by `xphd matdyn`.
- The exciton-momentum index is 0-based on the command line: `xphd generate iQ` takes an index over the full zone, and the archive is named with $i_Q + 1$, so $i_Q = 0$ gives `GI_ExcPh_Q0001.npz` (the point $\Gamma$).
- States and bands on the command line are 1-based (`-state 1` is the lowest). The band window printed by yambo is 1-based; the archive field `bse_bands` is a Python slice (for example `[6, 12]` means bands 7–12).
- The fine $\mathbf{q}$-list for `matdyn` is $360 \times 360$, row-major, in crystal coordinates: point $i \cdot 360 + j$ is $(i/360,\ j/360,\ 0)$, with a count line first. Since 360 is divisible by 6, 8, 12, 18 and 24, the fine mesh contains every source mesh we used.
- Phonon branches are ordered by energy, so a branch index changes character wherever two branches cross. Selection rules are always applied mode by mode, never by index.

## 3.2 Electron–hole pairing

**This is the most important convention in the code.** yambo stores the exciton envelope with the electron at the table index $\mathbf{k}$ and the hole at $\mathbf{k} + \mathbf{Q}$ (equivalently, the exciton at $-\mathbf{Q}$ in the convention "electron at $\mathbf{k}$, hole at $\mathbf{k} - \mathbf{Q}$"). `xphd` shifts the valence rotation matrix accordingly. Only this pairing gives $|\chi| = 1$ for every exciton, in 100% of states; the other three ways of attaching the shift give at most 35%. A wrong pairing produced a fictitious "mixed-parity ring" around $\Gamma$ in our early GaN results.

## 3.3 Mirror parity

- $\sigma_h$ leaves every in-plane wavevector unchanged, so in any layer whose point group contains $\sigma_h$ every non-degenerate state has an exact parity at every $\mathbf{k}$.
- *Electrons:* the diagonal of $D(\sigma_h)$ is $\pm 1$.
- *Excitons:* $\chi = \sum |A|^2 p_c p_v$, generalised to the full $D$-matrices, and it must be $\pm 1$ for every non-degenerate state.
- *Phonons:* $p = \sum_\kappa \langle \mathbf{e}_\kappa | M\, \mathbf{e}_{\pi(\kappa)} \rangle$ with $M = \mathrm{diag}(1, 1, -1)$ and $\pi$ the atom permutation of $\sigma_h$. For a planar layer $\pi$ is the identity and $p = 1 - 2w_z$, where $w_z$ is the out-of-plane weight.
- *Selection rule:* an even phonon connects excitons of equal parity and an odd phonon connects excitons of opposite parity. For emission we use the axis-covariant form $\Gamma_{\mathrm{exc}} \subset \Gamma_\nu \otimes \Gamma_{\mathrm{photon}}$.

# 4. Data files

| File | Written by | Contents | Read by |
|---|---|---|---|
| `Dmats.npy` | `xphd dmats` | Single-particle rotation matrices $(n_{\mathrm{sym}}, n_k, [1,] n_b, n_b)$ for the BSE band window | generate, parity, band-parity, irreps, dipoles |
| `GI_ExcPh_QXXXX.npz` | `xphd generate` | Archive for one $\mathbf{Q}$: `G_grid` $(n_q, n_{\mathrm{mod}}, n_{\mathrm{init}}, n_{\mathrm{final}})$ complex, in eV; `g2_grid`$= \lvert G\rvert^2$; electron and hole parts `Ge_grid`, `Gh_grid`; `E_n_grid` (states at $\mathbf{Q}$); `E_m_grid` (states at $\mathbf{Q}+\mathbf{q}$); `hw_grid`; `q_red`, `Q_red`, `mesh`; `bse_nv`, `bse_nc`, `bse_bands`; `elph_convention` | everything downstream |
| `hw_fine.npy` | `xphd matdyn` | Phonon energies (eV) on the $360 \times 360$ mesh, row-major | linewidth, sweep, bte |
| `matdyn.modes` | `run_matdyn.sh` | Eigenvectors on the fine mesh, with one header | mode labels, parity maps |
| `parity.npz` | `xphd parity` | `Q_red` (full zone), `chi` $(n_Q, n_{\mathrm{states}})$, `E`, `iQ_ibz` | selection-rule and the parity plots |
| `band_parity.npz` | `xphd band-parity` | `kf`, `E`, `p`, `off`, `defined`, `bands` (1-based) | band-parity plots |
| `mode_labels_elph.npy` | `labels_from_elph.py` | $(n_q, n_{\mathrm{mod}})$ phonon parity in the archive's own order: odd $\to 3$, even $\to 1$, undefined $\to -1$ | selection-rule |
| `exc_dipoles.npy` | `xphd dipoles` | Complex exciton dipoles $(3, n)$ of the $\mathbf{Q} = 0$ states of one BSE | chirality, helicity analysis |
| `lw_77K.npz` | `xphd sweep, unfold` | Linewidth field over the zone | bte-check |

| File | Written by | Contents | Read by |
|---|---|---|---|
| snaps_77K.npz | xphd bte -snapshots | $t$, $F$, $E$, Q_red, band, mesh, $T$ (plus channel data with -channels) | check_bte, valley_times, arpes |

## 5. Command reference

The subcommands are grouped by stage. -mesh is derived from the archive or the lattice when it is omitted. Most commands that read yambo take -save (or -savepath), -bse-dir and -dmats.

| Command | What it does | Typical call | Writes |
|---|---|---|---|
| matdyn-split | Split the fine q-list into chunks for concurrent matdyn.x runs | -n 360 -chunks 48 -prefix qlist | qlist_###.txt, chunk_index.txt |
| matdyn-merge | Merge chunked .modes/.freq outputs with single headers and summed counts | -chunks -modes-prefix -freq-prefix -o -freq -index | matdyn.modes, .freq |
| modes-check | Verify a merged matdyn.modes: counts, duplicates, $\Gamma$ acoustic residue, imaginary modes, out-of-plane weights | -modes -freq -n 360 | report |
| matdyn | Read the fine-mesh .freq, convert to eV, cross-check against an archive's hw_grid | <freq> -fine 360 -nbnd N -npz <archive> -o hw_fine.npy | hw_fine.npy |
| dmats | Rotation matrices for the BSE band window | -bse-dir -savepath -ndb-elph -nexc | Dmats.npy |
| ibz | List the irreducible $\mathbf{Q}$-points | -savepath | list |
| generate | One archive for full-zone index $i_Q$ (0-based) | iQ [outdir] -nexc -savepath -ndb-elph -bse-dir -dmats -mesh n n | GI_ExcPh_Q{iQ+1:04d}.npz |
| check-archive | Validate one archive: fields; $\lvert G\rvert^2 =$ g2; $G_e + G_h = G$; $E_m(0) = E_n$; acoustic sum rule; symmetry (time reversal and $C_3$ at $\Gamma$, excitonic space sums; OK $< 10^{-2}$, FAIL $> 5 \times 10^{-2}$) | <archive> [-deg-tol 0.015] [-detail] | READY / NOT READY |
| check | Older archive integrity checks | <archive> | report |
| verify-archives | All $N^2$ archives present and complete | -dir -n | report |
| linewidth | Golden-rule linewidth of every state at the archive's $\mathbf{Q}$, with broadening-free triangles on the refined mesh | <archive> -hw-fine -T ... -acoustic-cut 5e-4 -acoustic-model q2 -n-shell 1 [-per-mode] -o | text table (meV, fs; emission and absorption) |
| sweep | Linewidths at every $\mathbf{Q}$ in a set of archives | 'GI_ExcPh_Q*.npz' -hw-fine -T 77 -acoustic-model q2 -n-shell 1 -workers 8 -unfold SAVE -o | field npz |
| unfold | Symmetrise or expand an irreducible field onto the full zone | <field> -savepath -o | field npz |
| interference | Split the width into electron, hole and cross terms | <archive> -hw-fine -T | report |
| channels | Which final states and branches carry the width | <archive> -state -T -top | report |
| frohlich | Electron/hole Fröhlich cancellation at small $q$ | <archive> -nshell | report |
| selfenergy | $\Sigma(\omega)$, quasiparticle energy and spectral function (Kramers–Kronig) | <archive> -state -T -hw-fine -o | npz |
| lineshape | Full $\Sigma$ matrix and absorption lineshape (off-diagonal terms) | <archive> -T -hw-fine -bsdb ndb.BS_diago_Q1 -o | npz |
| localization, gauge | Real-space localisation and gauge-smoothness diagnostics | <archive> -hex <a in Å> | report |
| parity | $\chi(\sigma_h)$ of every state at every $\mathbf{Q}$ (irreducible set, unfolded by yambo's own map); with -manifold-tol also the $\sigma_h$ matrix inside each near-degenerate excitonic space | -save -bse-dir -dmats -nv -nstates [-manifold-tol 0.005] -o | parity.npz |

| Command | What it does | Typical call | Writes |
|---|---|---|---|
| band-parity | $\sigma_h$ parity of the electronic bands | -save -bse-dir -dmats -nv -path G M K G -o | band_parity.npz |
| irreps | Irreducible representations at high-symmetry points | -save -bse-dir -dmats -nv -points K M -nstates -deg-tol | table |
| selection-rule | Coupling split into allowed, forbidden and undefined by parity; maps. -rotate splits by the parity eigenstates of each near-degenerate excitonic space, so solver-mixed states are not excluded | -archive -parity -labels -state [-by-final] [-tol] [-rotate] | PDF and numbers |
| mode-labels | Branch characters (ZA, ..., LO) from matdyn eigenvectors; planar two-atom layers only (prefer labels_from_elph.py) | -modes -fine -mesh -masses -archive | mode_labels.npy |
| decompose-parity | Why a state has a fractional $\chi$ | -parity -nv -iq -state | report |
| check-parity-asr | Band mirror parity and acoustic-sum-rule residue | -labels -hw-fine -nv | report |
| validate-planar | Checks for a planar structure (planar only) | -masses -savepath | report |
| dipoles | Complex exciton dipoles from ndb.dipoles and one set of BSE eigenvectors, compared with yambo's residuals by excitonic space | -save -bse-dir -o [-deg-tol 0.005] | exc_dipoles.npy |
| chirality | Helicity and valley polarisation of a degenerate pair | -dipoles -states i j | report |
| coherence | $k$-space overlap (interference figure of merit) for a pair | -pair i j | report |
| arpes | Exciton photoemission $I(\mathbf{k}, E)$, thermal or time-resolved | -temperature T \| -snapshots -time | npz |
| bte | Real-time Boltzmann transport | 'GI_ExcPh_Q*.npz' -T 77 -refine 5 -hw-fine -inject optical -pump-center -pump-sigma -t-end -nt -workers -channels -snapshots | snapshots npz |
| bte-check | Do the transport out-rates equal the linewidths? | -archives -field -hw-fine -T | report |
| effective-mass | Parabolic fit to a band edge from bands.x output | bands.gnu -band -center | report |
| mott-wannier | 2D Mott–Wannier estimate of the bright–dark splitting | -Eg-dir -Eg-ind -mu-dir -mu-ind -alpha | report |

**Always pass** -acoustic-model q2 -n-shell 1 **to** linewidth **and** sweep. The default is none. At a band minimum the only decay channel is quasi-elastic acoustic absorption inside the first mesh cell, so without the q2 model the code returns the rate of escape out of the valley bottom instead of the width of the state. That is a useful quantity, but a different one (Sec. 10.5).

## 5.1 Scripts

Copy the script into your working folder, edit the SETTINGS block and run it.

| Script | Purpose | Key settings |
|---|---|---|
| labels_from_elph.py | Phonon parities in the electron–phonon database's own mode order (permutation formula; planar and non-planar) | ELPH, ARCHIVE, SAVE (or PERM), OUT |
| crossing_radii.py | Exciton parity contour (interpolated) and the crossing radius of a branch from out-of-plane to in-plane character | PARITY, MODES ($360 \times 360$), N_FINE, BRANCH |
| check_minimum_linewidth.py | Closed-form check of the linewidth at a band minimum (curvature, LA velocity, coupling) | ARCHIVE, HW_FINE, ALAT, MASSES, TEMPERATURES |
| compare_band_interp.py | Local-quadratic against Fourier band refinement, on held-out BSE points | ARCHIVE, BANDS |

| Script | Purpose | Key settings |
|---|---|---|
| `where_forbidden.py` | What carries any forbidden coupling: phonon swaps, solver-mixed near-degenerate states, or diffuse admixture $\varepsilon = (1-\lvert\chi\rvert)/2$; prints a verdict and the `-tol` that confirms it | ARCHIVE, PARITY, LABELS, STATE, TOL |
| `manifold_closure.py` | Are the near-degenerate excitonic spaces whole inside the archive's states, or does the state cut split them? | PARITY (more states than the archive), NARCHIVE |
| `window_closure.py` | Is the BSE band window closed under $\sigma_h$? Norms of $D(\sigma_h)$ band by band; a leaking band caps exciton parities below 1 | DMATS, SIGMA_H_OP, FIRST_BAND, NV |
| `star_uniformity.py` | Does the integrated coupling $T(\mathbf{q})$ agree across every star of $\mathbf{q}$? (Sec. 10.2) | ARCHIVE, SYM, INITIAL, N_FINAL |
| `irreps_check.py` | Irreducible representation of a chosen excitonic space at a high-symmetry point, from the rotation matrices | SAVE, BSE_DIR, DMATS, POINT, STATES |
| `phonon_irreps_K.py` | Irreducible representations of the phonons at $\mathbf{K}$ from the eigenvectors, for any choice of $C_3$ axis | MODES, POSITIONS, AXIS, K_POINT |
| `add_osc.py` | Add the $\mathbf{Q} = 0$ oscillator strengths (`osc_Q0`) that `bte -inject optical` needs to the $\Gamma$ archive | ARCHIVE, DIPOLES, POLARISATION |
| `check_bte.py` | Has the BTE settled, and where? Mean energy over time, and the share at the minimum's point and in each valley, against Boltzmann on the same states | BTE, T_BTE, T_CHECK, CENTERS, RADIUS |
| `valley_times.py` | Times at which each valley's population crosses 50, 90 and 99% | SNAPS, CENTERS, RADIUS |
| `exc_transitions.py` | Band transitions and electron valley weights of the lowest excitons at any $\mathbf{Q}$ | SAVE, BSE_DB, NSTATES, CENTERS |
| `final_state_window.py` | How close the final states come to the edge of the archive's state window, archive by archive | ARCHIVES, N_INITIAL |
| `slice_archive.py` | Copies of an archive with fewer states, to test convergence of a width with the number of final states | ARCHIVE, N_STATES, STATE, T |
| `run_matdyn.sh` | Chunked `matdyn.x` on the $360 \times 360$ list, with merged outputs | (shell variables) |
| `run_generate.sh` / `.ps1` | Every archive of the full zone with `xphd generate`, resumable; refuses to start unless the $\Gamma$ archive is READY | N, NEXC, SAVE, ELPH, BSE, DMATS |

# 6. Upstream settings

`xphd` trusts its inputs, so these settings must hold or nothing downstream means anything.

## 6.1 Quantum ESPRESSO: ground state and phonons

- Use separate directories for scf, phonons and nscf. The nscf overwrites `<prefix>.save`, and `ph.x` must read the scf ground state, so copy the scf save into each.
- Take the q-grid equal to the k-grid ($N \times N \times 1$) so that every $\mathbf{k}+\mathbf{q}$ lies on the mesh.
- With images (`ph.x -ni N`), each image works in its own `_ph{i}` directory, so `dvscf` and patterns are spread over them (gathered by `copy.sh`, below). A restart (`recover=.true.`) must keep the same process and pool layout.
- An error such as `diropn ... error opening ./_ph0/<prefix>.q_N/<prefix>.mixd97`, on many ranks at once, is a file-system problem rather than physics. Typical causes are a restart with a different layout, a node that cannot see `outdir`, or a full quota. Keep `outdir` on shared scratch, consider `reduce_io=.true.`, and split the $\mathbf{q}$-points with `start_q`/`last_q`.

## 6.2 q2r.x and matdyn.x on the $360 \times 360$ mesh

```
q2r.in
&input
  fildyn = '<prefix>.dyn'
  zasr   = 'crystal'
  flfrc  = '<prefix>.fc'
  loto_2d = .true.
/
```

```
matdyn.in  (namelist ONLY, ends with '/')
&input
  asr              = 'crystal'
  flfrc            = '<prefix>.fc'
  flfrq            = '<prefix>.freq'
  flvec            = 'matdyn.modes'
  loto_2d          = .true.
  q_in_cryst_coord = .true.
  q_in_band_form   = .false.
  nosym            = .true.
  dos              = .false.
```

```
/
```

- `loto_2d=.true.` in both programs: the three-dimensional non-analytic form adds a spurious LO splitting at $\Gamma$.
- `qlist.txt` holds the $360 \times 360$ points in row-major crystal coordinates with a count line (129 600 points).
- `run_matdyn.sh` (Slurm) runs `matdyn.x` in 54 chunks of 2 400 points and merges `matdyn.modes`, `.freq` and `.freq.gp` with single headers, summed counts and a monotonic path distance. It also checks the namelist's closing `/`, that the force constants exist, and every count. None of `matdyn`'s outputs survives a plain `cat`, because each chunk carries its own header and cumulative distances.
- `matdyn.x` is serial, so the parallelism is several chunks at once.

## 6.3 LetzElPhC

- Build `LELPH/ph.save` with `copy.sh`, run in the phonon folder. It gathers `dyn0..dynN`, `dvscf1..dvscfN`, `patterns.N.xml`, `tensors.xml`, `data-file-schema.xml` and the pseudopotentials. It searches every `_ph*` image (LetzElPhC's own `-pp` reads patterns only from `_ph0`), stops if one $\mathbf{q}$'s `dvscf` differs between images (a split $\mathbf{q}$ leaves partial files), and flags `dvscf` files of unequal size.
- In `lelphc.in` set `ph_save_dir = ph.save`. The default is `ph_save`, the key is case-sensitive, and a misspelling is silently ignored, which then shows up as `Unable to open the dyn0 file`.
- **Same SAVE rule.** LetzElPhC and the BSE must read the same yambo `SAVE`; the exciton–phonon elements combine both in one wavefunction gauge. If you redo the nscf, redo both.

## 6.4 yambo: GW and BSE

- For a two-dimensional layer use `CUTGeo= "slab z"` with the RIM (`rim_cut`). In GW, the Bruneval–Gonze terminator (`GTermKind="BG"`, `XTermKind="BG"`) speeds up band convergence; leave `VXCRLvcs` at its default.
- BSE: `BSEBands` is the band window (it fixes `-nv`); `BSEQptR` covers all irreducible $\mathbf{Q}$; `BSSmod= "d"` is full diagonalisation. The matrices are small (5 184 for GaN at $24 \times 24$) and the solver cannot fail to converge.
- Check the quasiparticle correction in the BSE report. In the build we used (5.2.0 r22427, `devel-excph`), we confirmed that the report's `[K+QP] Direct Gap` equals the `[X]` gap plus the GW correction. Make the same check on your build before trusting any energy.
- "Maximum Residual Value" in the sorted-exciton files is the normalisation of the oscillator strengths (yambo's "residuals" are the numerators of the spectrum). It is not a convergence measure.
- At $\mathbf{Q} = 0$ the long-range exchange is evaluated along one direction, which splits each $E'$ doublet into a longitudinal and a transverse member (in GaN, 2.8 meV for the first bright pair and 10 meV for the second). Each member alone is not $C_3$-symmetric; only the pair is.

## 6.5 From databases to archives

### 1. Rotation matrices

```
xphd dmats --bse-dir <BSE_DIR> --savepath <SAVE> --ndb-elph <ELPH> --nexc <n>
```

| | |
|---|---|
| **What it does** | Builds the single-particle rotation matrices $D$ for every symmetry operation and $\mathbf{k}$, restricted to the BSE band window. |
| **Needs** | `SAVE`, the BSE databases, `ndb.elph`. |
| **Writes** | `Dmats.npy` with shape $(n_{\mathrm{sym}}, n_k, [1,]n_b, n_b)$. |
| **Check** | The matrices depend on the `SAVE` and the band window, not on the BSE solution, so they stay valid if only the BSE is redone. |

### 2. The $\Gamma$ archive, then the check

```
xphd generate 0 <OUT> --nexc <n> --savepath <SAVE> --ndb-elph <ELPH> --bse-dir <BSE_DIR> \
      --dmats Dmats.npy --mesh N N
xphd check-archive <OUT>/GI_ExcPh_Q0001.npz [--detail]
```

**What it does** Computes $G(\mathbf{Q}=0,\mathbf{q})$ for every $\mathbf{q}$ of the mesh, every branch and every pair of states. Excitons at $\mathbf{Q}+\mathbf{q}$ are rotated from the irreducible BSE with the $D$-matrices. The check validates fields, identities and symmetry.

**Needs** `Dmats.npy`, `SAVE`, `ndb.elph`, the BSE.

**Writes** `GI_ExcPh_Q0001.npz`.

**Check** READY; time reversal and $C_3$ both at the $10^{-3}$ level (GaN: $1.3\times10^{-3}$ and $3.3\times10^{-3}$). One operation far above the other means a rotation fault, unless `-detail` points to a split doublet (then raise `-deg-tol`).

**3. Every archive of the full zone** (needed only for sweeps and transport)

```
bash run_generate.sh                # Slurm; set N, NEXC, SAVE, ELPH, BSE, DMATS
xphd verify-archives --dir <OUT> --n <N*N>
```

**What it does** Runs `xphd generate` for $i_Q = 1\ldots N^2-1$, several at once (it is memory-bound) and resumable. It refuses to start unless the $\Gamma$ archive is READY.

**Needs** A READY $\Gamma$ archive.

**Writes** $N^2$ archives.

**Check** `verify-archives`: all present and complete.

# 7. Worked example: planar monolayer GaN

GaN is the case where the symmetry matters most, because its lowest exciton is momentum-dark. The parameters below are the ones we used; each step shows the command, what it needs, what it writes, and the numbers we obtained, so that you can compare your own run.

| Parameter | Value |
|---|---|
| **Structure** | planar $D_{3h}$ honeycomb, 2 atoms (Ga, N) |
| **Mesh** | $24\times24$ (576 k/q-points, 61 irreducible); fine mesh $360\times360$ |
| **BSE** | bands 7–12 (`bse_bands [6 12]`), `-nv 3`, 15 states per $\mathbf{Q}$ |
| **Phonons** | 6 branches; masses 69.723 (Ga), 14.007 (N); $\pi$ = identity |
| **Key energies** | lowest exciton at $\mathbf{K}$: 3.4290 eV (dark, $\sigma_h$-odd, A″); bright E′ doublet at $\Gamma$: 3.8143 and 3.8171 eV; $G_0W_0$ gap 4.78 eV indirect (K→ $\Gamma$), 4.91 eV direct |
| **Archive of $\mathbf{K}$** | `GI_ExcPh_Q0575.npz` |

## 7.1 Commands, in order

**1. Fine-mesh phonons**

```
xphd matdyn gan.freq --fine 360 --nbnd 6 --npz GI_ExcPh_Q0001.npz -o hw_fine.npy
```

**What it does** Reads the merged $360\times360$ frequencies, converts them to eV and checks that they reproduce the archive's `hw_grid` where the meshes coincide.

**Needs** `gan.freq` from `run_matdyn.sh`.

**Writes** `hw_fine.npy`.

**Check** Agreement with the source mesh at the $\mu$eV level.

**2. Linewidth of the bright exciton ($\Gamma$)**

```
xphd linewidth GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
      --acoustic-model q2 --n-shell 1 --per-mode -o G.txt
```

**What it does** Golden-rule width of every state at $\Gamma$. The energy-conserving $\delta$ is integrated on $360\times360$ triangles, exciton energies are refined by local quadratics, and the acoustic coupling inside the first cell follows $|G|^2\,2\hbar\omega = Cq^2$.

**Needs** The $\Gamma$ archive and `hw_fine.npy`.

**Writes** `G.txt` (meV, fs, emission and absorption, per branch).

**Check** State 1: 36.7 meV at 5 K (99.8% emission), 51.9 meV at 77 K (12.7 fs), 150.9 meV at 300 K (43% absorption); state 2: 56.8 meV at 77 K.

### 3. The $\mathbf{K}$ minimum: width and escape

```
xphd linewidth GI_ExcPh_Q0575.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
      --acoustic-model q2 --n-shell 1 --per-mode -o K.txt
xphd linewidth GI_ExcPh_Q0575.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 -o K_none.txt
```

**What it does** With `q2`, the width of the minimum, from quasi-elastic LA absorption inside the valley. Without it (the default, `none`), the rate of scattering beyond the first mesh cell, which is the escape from the valley bottom.

**Needs** The $\mathbf{K}$ archive and `hw_fine.npy`.

**Writes** `K.txt`, `K_none.txt`.

**Check** `K.txt`, state 1: 0.087, 10.4 (63 fs) and 50.5 meV (13 fs) at 5, 77 and 300 K. `K_none.txt`: 0.0437 meV at 77 K, i.e. about 15 ps.

### 4. Closed-form check of the $\mathbf{K}$ minimum

```
python check_minimum_linewidth.py     # ARCHIVE = GI_ExcPh_Q0575.npz, ALAT, MASSES
```

**What it does** Evaluates $\Gamma = (4\pi^2/\sqrt{3})\, C\, N(\hbar\omega^*)/b^2$ with $\hbar\omega^* = (\hbar c)^2/b$ from the BSE curvature, the LA velocity and the fitted coupling, and compares it with the code (Sec. 10.5).

**Needs** The $\mathbf{K}$ archive and `hw_fine.npy`.

**Writes** A report.

**Check** $b = 12.44$ eV $|\mathbf{b}|^{-2}$ and $\hbar\omega^* = 1.49$ meV; the closed form gives 10.6–11.4 meV at 77 K against 10.4 meV from the code.

### 5. Linewidths over the zone

```
xphd sweep 'GI_ExcPh_Q*.npz' --hw-fine hw_fine.npy --T 77 --acoustic-cut 5e-4 --acoustic-model q2 \
      --n-shell 1 --workers 8 --unfold SAVE -o lw_77K.npz
```

**What it does** Linewidths at every irreducible $\mathbf{Q}$, unfolded onto the full zone.

**Needs** The irreducible archives (or all), `hw_fine.npy`, `SAVE`.

**Writes** `lw_77K.npz`.

**Check** Repeat at 300 K if you want a map at that temperature.

### 6. Exciton parity across the zone

```
xphd parity --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv 3 --nstates 15 -o parity.npz
```

**What it does** $\chi(\sigma_h)$ of every state at every irreducible $\mathbf{Q}$, unfolded with yambo's own map.

**Needs** `SAVE`, the BSE at all $\mathbf{Q}$, `Dmats.npy`.

**Writes** `parity.npz`.

**Check** Every state is $\pm 1$; the lowest band is even inside a contour around $\Gamma$ and odd outside it (including $\mathbf{K}$ and $\mathbf{M}$).

### 7. Phonon parities in the archive's mode order

```
python labels_from_elph.py        # ELPH, ARCHIVE = GI_ExcPh_Q0001.npz, SAVE
```

**What it does** Parity of every $(\mathbf{q}, \text{mode})$ from the electron–phonon database's own eigenvectors. (`matdyn` orders nearly degenerate modes differently at about 5% of the points, which is why we do not use its labels.)

**Needs** `ndb.elph`, the archive, `SAVE`.

**Writes** `mode_labels_elph.npy`.

**Check** It reports a planar layer, and the frequencies match the archive's (same order).

### 8. The selection rule, quantified

```
xphd selection-rule --archive GI_ExcPh_Q0001.npz --parity parity.npz --labels mode_labels_elph.npy \
      --state 1 --by-final
```

**What it does** Splits the coupling from the initial state into allowed, forbidden and undefined by parity.

**Needs** The archive, `parity.npz`, the labels.

**Writes** Numbers and maps.

**Check** Forbidden/allowed $= 5 \times 10^{-7}$, i.e. zero to numerical precision.

### 9. Labels at $\mathrm{K}$ and $\mathrm{M}$; band parity

```
xphd irreps --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv 3 --points K M --nstates 4
xphd band-parity --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv 3 --path G M K G -o band_parity.npz
```

**What it does** Irreducible representations of the lowest states, and the parity of the electronic bands.

**Needs** `SAVE`, the BSE, `Dmats.npy`.

**Writes** Tables and `band_parity.npz`.

**Check** The lowest exciton at $\mathbf{K}$ is A″ (odd). The valence maximum at $\mathbf{K}$ is N $p_z$ (odd), and the conduction minimum at $\Gamma$ is Ga $s$ (even).

### 10. Parity contour

```
python crossing_radii.py          # PARITY = parity.npz, MODES = 360x360 matdyn.modes, BRANCH = 4
```

**What it does** The interpolated crossing of the lowest even and odd exciton bands, and the radius at which a chosen phonon branch changes from out-of-plane to in-plane character.

**Needs** `parity.npz`, `matdyn.modes`.

**Writes** The radii along $\Gamma$–K and $\Gamma$–M.

**Check** The contour lies at $0.256\,|\mathbf{b}|$ along $\Gamma$–K and $0.265\,|\mathbf{b}|$ along $\Gamma$–M.

### 11. Exciton dipoles and the helicity-resolved sideband

```
xphd dipoles --save <SAVE> --bse-dir <BSE_DIR> -o exc_dipoles.npy --deg-tol 0.005
python helicity_pl.py             # ARCHIVE = GI_ExcPh_Q0001.npz, DIPOLES = exc_dipoles.npy
```

**What it does** Complex $x, y, z$ dipoles of the $\mathbf{Q} = 0$ states from one BSE; then the phonon-assisted luminescence resolved by circular polarisation and by valley.

**Needs** `SAVE`, the BSE (`ndb.dipoles`), the $\Gamma$ archive.

**Writes** `exc_dipoles.npy` and the helicity-resolved spectra.

**Check** yambo's residuals agree per excitonic space. The satellite sits at 3.4128 eV, 16.2 meV below the dark exciton; $P = -1.0000$ at $\mathbf{K}$ and $+1.0000$ at $\mathbf{K}'$; the sum rule holds to $3.7 \times 10^{-16}$; the valley intensities agree to 0.12% (ratio 0.9988).

### 12. Real-time transport

```
xphd bte 'GI_ExcPh_Q*.npz' --T 77 --refine 5 --hw-fine hw_fine.npy --inject optical \
      --pump-center <E_bright> --pump-sigma 0.018 --t-end 2000 --nt 200 --workers 14 \
      --channels --snapshots snaps_77K.npz
python check_bte.py ; python valley_times.py
xphd bte-check --archives 'GI_ExcPh_Q*.npz' --field lw_77K.npz --hw-fine hw_fine.npy --T 77
```

**What it does** Propagates the populations of all states at all $\mathbf{Q}$ with the full rate matrix, with detailed balance symmetrised and the rates refined by row rescaling (Sec. 10.7).

**Needs** All 576 archives and `hw_fine.npy`.

**Writes** Snapshots and diagnostics.

**Check** The population relaxes to 4.26 meV above the minimum, against a Boltzmann value of 4.09 meV at 77 K. The $\mathbf{K}$ valleys hold 35% of the population at 30 fs, 50% at 41 fs, 90% at 98 fs and 99% at 172 fs.

#### 13. Diagnostics, as needed

```
xphd channels GI_ExcPh_Q0001.npz --state 1 --T 77
xphd interference GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 77
xphd frohlich GI_ExcPh_Q0001.npz
xphd lineshape GI_ExcPh_Q0001.npz --T 77 --hw-fine hw_fine.npy --bsdb <BSE_DIR>/ndb.BS_diago_Q1 -o eps2.npz
```

**What it does** Where the width comes from; electron/hole interference; Fröhlich cancellation; off-diagonal self-energy.

**Needs** The archives.

**Writes** Reports.

**Check** For the polar branches, nothing is emitted from the bright $\Gamma$ state (parity); check the absorption at 300 K in the per-mode columns.

## 8. Worked example: monolayer hBN, the bright-minimum contrast

hBN is the natural comparison for GaN because it has the same lattice symmetry but the opposite exciton ordering: its lowest exciton is at $\Gamma$ and is bright, while the $\mathbf{K}$ exciton lies about 150 meV above it. The upstream steps are identical, so only the points that differ are given.

| Parameter | Value |
|---|---|
| **Structure** | planar $D_{3h}$, 2 atoms (B, N); $a$ from the relaxed `scf.out` ($\approx 2.50$ Å) |
| **Mesh** | $24 \times 24$ (576), fine mesh $360 \times 360$ |
| **BSE** | `bse_bands [1 7]`, `-nv 3`, 15 states |
| **Phonons** | 6 branches; masses 10.81 (B), 14.007 (N); $\pi$ = identity |
| **Known features** | lowest exciton at $\Gamma$ at 5.475 eV, bright, with the E′ doublet split by 0.2 meV; photoluminescence shows LO/TO sidebands of the bright exciton |

**Mesh convergence.** Going from $18 \times 18$ to $24 \times 24$ lowers the hBN exciton by 200–280 meV, so absolute exciton energies are not converged on these meshes. The symmetry statements, which do not depend on the absolute energy, are unaffected.

### 8.1 Commands, in order

#### 1. Check that the BSE energies went in correctly

```
grep -E "\[K\+QP\] Direct Gap|\[X\] Direct Gap" <BSE folder>/r-*
```

**What it does** Confirms that the quasiparticle correction was added to the Kohn–Sham energies (Sec. 10 and the yambo notes above).

**Needs** The BSE report.

**Writes** —

**Check** The [K+QP] gap equals the [X] gap plus the GW correction.

#### 2. Archives

```
xphd dmats --bse-dir ../BSE/output_all --savepath ../phonons/SAVE --ndb-elph ../phonons/ndb.elph --nexc 15
xphd generate 0 archives --nexc 15 --savepath ../phonons/SAVE --ndb-elph ../phonons/ndb.elph \
      --bse-dir ../BSE/output_all --dmats Dmats.npy --mesh 24 24
xphd check-archive archives/GI_ExcPh_Q0001.npz
```

**What it does** As in Sec. 6.5.

**Needs** `SAVE`, `ndb.elph`, the BSE.

**Writes** `Dmats.npy` and the archives.

**Check** READY; symmetry at the $10^{-3}$ level.

### 3. Parity, labels and the selection rule

```
xphd parity --save ../phonons/SAVE --bse-dir ../BSE/output_all --dmats Dmats.npy --nv 3 --nstates 15 -o parity.
   npz
python labels_from_elph.py        # ELPH = ../phonons/ndb.elph, ARCHIVE, SAVE = ../phonons/SAVE
xphd selection-rule --archive archives/GI_ExcPh_Q0001.npz --parity parity.npz \
      --labels mode_labels_elph.npy --state 1 --by-final
```

**What it does** As for GaN.

**Needs** As for GaN.

**Writes** `parity.npz`, the labels, numbers.

**Check** Forbidden/allowed is near zero, and the lowest exciton is even at $\Gamma$.

### 4. Labels and band parity

```
xphd irreps --save ../phonons/SAVE --bse-dir ../BSE/output_all --dmats Dmats.npy --nv 3 --points K M --nstates 4
xphd band-parity --save ../phonons/SAVE --bse-dir ../BSE/output_all --dmats Dmats.npy --nv 3 -o band_parity.npz
```

**What it does** Irreducible representations at $\mathbf{K}$ and $\mathbf{M}$, and the parity of the band edges (both $\pi$-like at $\mathbf{K}$: class I, Sec. 10.3).

**Needs** As above.

**Writes** Tables and `band_parity.npz`.

**Check** Both band edges are odd ($p_z$) at $\mathbf{K}$; the lowest exciton is even and bright.

### 5. Negative control for the helicity

```
xphd dipoles --save ../phonons/SAVE --bse-dir ../BSE/output_all -o exc_dipoles.npy --deg-tol 0.005
python helicity_pl.py             # hBN Gamma archive
```

**What it does** A bright minimum at $\Gamma$ should have no valley-locked sideband, so hBN is the control for the GaN result.

**Needs** `SAVE`, the BSE, the $\Gamma$ archive.

**Writes** The helicity-resolved spectra.

**Check** $|P|$ near zero, and no valley satellite.

### 6. Linewidths (optional)

```
xphd matdyn bn.freq --fine 360 --nbnd 6 --npz archives/GI_ExcPh_Q0001.npz -o hw_fine.npy
xphd linewidth archives/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
      --acoustic-model q2 --n-shell 1 --per-mode -o Gamma.txt
python check_minimum_linewidth.py   # ARCHIVE = Gamma archive, ALAT from scf, MASSES = [10.81, 14.007]
```

**What it does** The bright $\Gamma$ state is hBN's minimum, so its width is set by in-valley acoustic absorption (the `q2` model is essential).

**Needs** The $\Gamma$ archive and `hw_fine.npy`.

**Writes** `Gamma.txt`.

**Check** The closed form agrees with the code.

# 9. Pitfalls we found, and what fixed them

Each row is a problem we met while building the package, with the cause and the fix. If your numbers look wrong, check this table first.

| Symptom | Cause | Fix |
|---|---|---|
| A ring of fractional parity (0.26–0.38 $\|\mathbf{b}\|$) in GaN | Wrong electron–hole pairing in the exciton rotation | The valence matrix is shifted (`rep_matrix_Q`); every state is then $\pm 1$ |
| About 1.6% "forbidden" coupling | Phonon labels taken from `matdyn`, which orders modes differently from the electron–phonon database at $\sim 5\%$ of points | `labels_from_elph.py`, which gives $5 \times 10^{-7}$ |
| $\mathbf{K}$ linewidth 34% off, with spurious anisotropy | Global Fourier interpolation of energy-ordered exciton bands rings at crossings | Local quadratic refinement (default `-band-interp local`) |
| $\mathbf{K}$ minimum width $\sim 0.04$ meV instead of 10 meV | The acoustic channel inside the first cell is missing | `-acoustic-model q2 -n-shell 1` (the default is `none`) |
| Old script: "could not be unfolded" | Hard-coded table of irreducible $\mathbf{Q}$ from one material | `xphd parity`, which uses yambo's own map |
| yambopy "open-shell" error, or a shifted conduction block when reading `ndb.dipoles` | yambopy miscounts `DipBandsAll` bands when the first band is not 1 | `xphd dipoles` reads the full file and selects the window |
| Helicity-resolved PL identical for $\sigma^{\pm}$ | Valleys summed (time reversal) and/or dipoles taken from two BSE runs | Use one BSE's complex $x, y$ dipoles and valley-resolved sums |
| "Maximum Residual Value $= 0.2$" read as non-convergence | It normalises the oscillator strengths | No action needed; the diagonalisation is full anyway |
| `ph.x diropn` errors on many ranks | Restart with another layout, node visibility or quota | Keep the same layout on `recover`, use a shared `outdir`, and use `start_q`/`last_q` |
| LetzElPhC: "Unable to open the dyn0 file" | `ph_save_dir` not set (the default is `ph_save`) | `ph_save_dir = ph.save`, in lower case |
| Phonon parities wrong for a non-planar layer | The out-of-plane weight assumes every atom sits on the mirror plane | Use the permutation formula (`PERM` in the scripts); `mode-labels` refuses structures that are not planar two-atom layers |
| A small forbidden residue in states with $\|\chi\| \approx 0.9$ | A near-degenerate pair of opposite parity returned mixed by the solver | `-tol 0.96` confirms it; `-rotate` uses the excitonic space's eigenstates and excludes nothing |
| GaN $C_3$ check FAIL (0.20) while time reversal is OK | A doublet split by 10 meV through the long-range exchange; its members are individually anisotropic at small $q$ | `-deg-tol` defaults to 15 meV; `-detail` names split partners |
| `matdyn` outputs corrupted by `cat` | Headers and cumulative distances in every chunk | `run_matdyn.sh` / `matdyn-merge` |
| Emission rule not invariant to the choice of axis | The conjugated form $\Gamma_\nu \subset \Gamma_{\rm exc}^{*} \otimes \Gamma_{\rm photon}$ renames helicities | Use $\Gamma_{\rm exc} \subset \Gamma_\nu \otimes \Gamma_{\rm photon}$ |

# 10. Methods and derivations

This part explains what the code computes. For each quantity we state what it is, derive the working formula, say how it is evaluated on the mesh, and give the checks that tell you it is right, with the numbers we obtained for GaN and hBN.

## 10.1 Notation

**Lattice.** The hexagonal reciprocal basis $\mathbf{b}_1, \mathbf{b}_2$ encloses 60°. Reduced coordinates $(q_1, q_2)$ give $\mathbf{q} = q_1\mathbf{b}_1 + q_2\mathbf{b}_2$, and lengths are quoted in units of $|\mathbf{b}| = |\mathbf{b}_1|$. In these units the Brillouin zone has area $A_{\rm BZ} = \sqrt{3}/2$, $\mathbf{K} = (1/3, 1/3)$, $\mathbf{K}' = (2/3, 2/3)$, and the $\mathbf{M}$ points are the edge midpoints, for example $(1/2, 0)$. The source mesh is $N \times N$ with $N = 24$ and spacing $h = 1/N$; the refined mesh used for integration is $360 \times 360$.

**Electron–phonon elements.** These are computed by LetzElPhC [1] in the yambo convention,

$$g_{n'n\nu}(\mathbf{k}, \mathbf{q}) = \langle \psi_{n'\mathbf{k}} | \Delta_{\nu\mathbf{q}} V | \psi_{n,\mathbf{k}-\mathbf{q}} \rangle, \tag{1}$$

and converted to the standard convention before projection. The phonon displacement normalisation is that of DFPT [2, 3].

**Exciton envelopes.** The BSE is solved with yambo [4] at every irreducible $\mathbf{Q}$, with full diagonalisation. We write $A_{vc\mathbf{k}}^{S\mathbf{Q}}$ with the electron at $\mathbf{k}$ and the hole at $\mathbf{k} - \mathbf{Q}$. yambo's BSE tables store the electron at the table index $\mathbf{k}$ and the hole at $\mathbf{k} + \mathbf{Q}$, which is the exciton at $-\mathbf{Q}$ in this convention. Time reversal gives the excitons at $\mathbf{Q}$ and $-\mathbf{Q}$ the same energies

and parities, so nothing physical changes, but every formula that pairs an electron with a hole must use the stored pairing (Sec. 10.3).

**Archives.** An archive `GI_ExcPh_QXXXX.npz` stores, for one exciton momentum $\mathbf{Q}$, the complex amplitudes

$$G[\mathbf{q},\nu,\lambda,\beta] = \langle \beta, \mathbf{Q}+\mathbf{q}|\partial V_\nu|\lambda,\mathbf{Q}\rangle = \mathcal{G}_{\beta\lambda\nu}(\mathbf{Q},\mathbf{q}) \tag{2}$$

in eV, for every $\mathbf{q}$ of the source mesh, every branch $\nu$, initial state $\lambda$ at $\mathbf{Q}$ and final state $\beta$ at $\mathbf{Q}+\mathbf{q}$, together with $|\mathcal{G}|^2$, its electron and hole parts, and the energies. The final envelope is the conjugated one, as in yambopy [5].

## 10.2 Exciton–phonon matrix elements

**Definition.** The exciton–phonon matrix element is the sum of a hole term and an electron term, in each of which one carrier scatters while the other keeps its state [6, 7]:

$$\begin{aligned}\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q},\mathbf{q}) = &\sum_{v,v',c,c',\mathbf{k}} A^{v,c,\mathbf{k}}_{\lambda,\mathbf{Q}}\left[g_{vv',\nu}(\mathbf{k}-\mathbf{Q},\mathbf{q})\,\delta_{c,c'}\right]\left(A^{v',c',\mathbf{k}}_{\beta,\mathbf{Q}+\mathbf{q}}\right)^* \\ &- \sum_{v,v',c,c',\mathbf{k}} A^{v,c,\mathbf{k}}_{\lambda,\mathbf{Q}}\left[g^*_{c'c,\nu}(\mathbf{k}+\mathbf{q},\mathbf{q})\,\delta_{v,v'}\right]\left(A^{v',c',\mathbf{k}+\mathbf{q}}_{\beta,\mathbf{Q}+\mathbf{q}}\right)^*.\end{aligned} \tag{3}$$

**Gauge.** Both factors in Eq. (3) are built from the same Kohn–Sham wavefunctions (one yambo `SAVE` for LetzElPhC and the BSE), so they share one single-particle gauge. Two freedoms remain on the exciton side. *Overall phases:* replacing $A^\lambda \to e^{i\theta_\lambda}A^\lambda$ and $A^\beta \to e^{i\theta_\beta}A^\beta$ multiplies $\mathcal{G}_{\beta\lambda\nu}$ by $e^{i(\theta_\lambda-\theta_\beta)}$, which cancels in $|\mathcal{G}_{\beta\lambda\nu}|^2$. *Degenerate excitonic spaces:* within a degenerate excitonic space $\mathcal{M}$ the solver's basis is arbitrary, $|\beta'\rangle = \sum_{\beta\in\mathcal{M}} U_{\beta\beta'}|\beta\rangle$ with $U$ unitary. Then $\sum_{\beta'\in\mathcal{M}}|\mathcal{G}_{\beta'\lambda\nu}|^2 = \sum_{\beta\in\mathcal{M}}|\mathcal{G}_{\beta\lambda\nu}|^2$, and likewise for the initial excitonic space. Only sums over complete excitonic spaces are basis-independent, and every invariant used below is such a sum.

**Excitons on the full zone.** The BSE is solved only at the irreducible momenta (61 for a $24\times24$ mesh). The exciton at any other momentum is the image of its irreducible parent under the operation $R$ that maps one to the other [8]:

$$A^{S,R\mathbf{Q}}_{vc,R\mathbf{k}} = \sum_{v'c'} D^c_{cc'}(R;\mathbf{k})\,A^{S\mathbf{Q}}_{v'c',\mathbf{k}}\left[D^v_{vv'}(R;\mathbf{k}-\mathbf{Q})\right]^*, \tag{4}$$

where $D^n(R;\mathbf{k})$ are the single-particle rotation matrices in the BSE band window, computed by LetzElPhC. For an operation that includes time reversal, the parent envelope is complex conjugated first. The point is not to fix phases, which cancel anyway, but to make every member of a star the exact symmetry image of the same parent. A mapping that does not do so breaks the symmetry of $|\mathcal{G}|^2$.

**Symmetry tests.** *Star uniformity* (`star_uniformity.py`). The integrated strength $T(\mathbf{q}) = \sum_{\nu\lambda\beta}|\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}=0,\mathbf{q})|^2$, summed over all branches and complete excitonic spaces, must be identical for every member of a star [9–11]. For GaN the ratio $T_{\max}/T_{\min}$ within a star is 1.0001 for the reconstructed full zone, against 2.25 for an earlier output whose excitons at the star members were not constructed as images of a common parent.

*Excitonic-space-resolved test* (`xphd check-archive`). For an archive at a time-reversal-invariant $\mathbf{Q}$, define for each $\mathbf{q}$ $S_{abc}(\mathbf{q}) = \sum_{\nu\in a}\sum_{\lambda\in b}\sum_{\beta\in c}|\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q},\mathbf{q})|^2$ over phonon spaces $a$ (branches within 0.1 meV), initial excitonic spaces $b$ and final excitonic spaces $c$ (states within a tolerance, default 15 meV). Symmetry requires $S(\mathbf{q}) = S(g\mathbf{q})$ for $g =$ time reversal ($\mathbf{q}\to-\mathbf{q}$) and for $g = C_3$. The reported deviation is

$$L_1 = \frac{\sum_{\mathbf{q}}\sum_{abc}|S_{abc}(\mathbf{q}) - S_{abc}(g\mathbf{q})|}{\frac{1}{2}\sum_{\mathbf{q}}\sum_{abc}\left[S_{abc}(\mathbf{q}) + S_{abc}(g\mathbf{q})\right]}, \tag{5}$$

with OK below $10^{-2}$ and FAIL above $5\times10^{-2}$. The $C_3$ matrix in reduced coordinates depends on the angle between $\mathbf{b}_1$ and $\mathbf{b}_2$: $(q_1,q_2)\to(-q_1-q_2,q_1)$ for 60° and $(-q_2,q_1-q_2)$ for 120°, and the code takes the one that leaves the exciton energies invariant. For GaN the result is $1.3\times10^{-3}$ for time reversal and $3.3\times10^{-3}$ for $C_3$. A rotation fault shows as one operation far above the other.

*Exchange-split doublets.* At $\mathbf{Q}=0$ the BSE evaluates the long-range exchange along one direction $\hat{\mathbf{e}}$, splitting each E′ doublet into members polarised parallel and perpendicular to $\hat{\mathbf{e}}$. For a member with real in-plane dipole $\mathbf{d}$, the coupling to small-$\mathbf{q}$ polar phonons carries the factor $|\hat{\mathbf{q}}\cdot\mathbf{d}|^2$. It is unchanged by $\mathbf{q}\to-\mathbf{q}$ but not by $C_3$, which rotates $\hat{\mathbf{q}}$ relative to the fixed $\mathbf{d}$. Summed over the doublet, $|\hat{\mathbf{q}}\cdot\mathbf{d}_\parallel|^2 + |\hat{\mathbf{q}}\cdot\mathbf{d}_\perp|^2 = |\mathbf{d}|^2$ is isotropic. With 5 meV excitonic spaces the second GaN doublet is split in two and the $C_3$ test fails ($L_1 = 0.20$, all of it in that doublet and the LO branch at the smallest $|\mathbf{q}|$); grouping it restores $3.3\times10^{-3}$.

## 10.3 Mirror parity and selection rules

**The horizontal mirror.** $\sigma_h$ maps $(x,y,z)\to(x,y,-z)$ and leaves every in-plane wavevector unchanged, so it belongs to the little group of every $\mathbf{Q}$ and commutes with the Kohn–Sham, phonon and BSE Hamiltonians at every point of the

zone. Every non-degenerate state therefore carries an exact parity across the whole zone. In a planar layer every atom lies in the mirror plane and maps onto itself.

**Electrons.** For spinless states the diagonal of $D(\sigma_h;\mathbf{k})$ gives $p_n(\mathbf{k}) = \langle\psi_{n\mathbf{k}}|\sigma_h|\psi_{n\mathbf{k}}\rangle = \pm 1$.

**Phonons.** $\sigma_h$ acts on a displacement pattern $\mathbf{e} = (\mathbf{e}_\kappa)$ as $(\sigma_h\mathbf{e})_\kappa = M\,\mathbf{e}_{\pi(\kappa)}$, with $M = \mathrm{diag}(1,1,-1)$ and $\pi$ the atom permutation (an involution). The parity of a normalised mode is

$$p_{\nu\mathbf{q}} = \langle\mathbf{e}_{\nu\mathbf{q}}|\sigma_h\mathbf{e}_{\nu\mathbf{q}}\rangle = \sum_\kappa \mathbf{e}^\dagger_{\nu\mathbf{q},\kappa}\, M\, \mathbf{e}_{\nu\mathbf{q},\pi(\kappa)} = \pm 1. \tag{6}$$

For a planar layer $\pi = \mathrm{id}$ and $p = 1 - 2w_z$ with $w_z = \sum_\kappa |e_{\kappa z}|^2$, so out-of-plane modes are odd. For a layer in which the mirror swaps atoms, this shortcut fails, because a mode can move along $z$ and still be even; Eq. (6) with the right permutation gives $\pm 1$ to machine precision, which is the reason the code takes the permutation as an input. The eigenvectors are taken from the electron–phonon database itself, in the order of the couplings: branches labelled from a separate `matdyn.x` run, ordered by its own frequencies, disagreed at 5% of the points in GaN.

**Excitons.** Applying a rotation $R$ to exciton $T$ and projecting on $S$,

$$D_{ST}(R) = \sum_{\mathbf{k}} \sum_{vc,v'c'} \left[A^S_{vc\mathbf{k}}\right]^* D^c_{cc'}(R;\mathbf{k})\, A^T_{v'c',R^{-1}\mathbf{k}} \left[D^v_{vv'}(R;\mathbf{k}-\mathbf{Q})\right]^*. \tag{7}$$

For $\sigma_h$, $R^{-1}\mathbf{k} = \mathbf{k}$, and for non-degenerate bands

$$\chi_S \equiv D_{SS}(\sigma_h) = \sum_{vc\mathbf{k}} |A^S_{vc\mathbf{k}}|^2\, p_c(\mathbf{k})\, p^*_v(\mathbf{k}-\mathbf{Q}), \tag{8}$$

which is real and equal to $\pm 1$. Using yambo's storage, the valence matrix must be taken at the hole's wavevector, $\mathbf{k}+\mathbf{Q}$ in table indices. Of the four ways of attaching the shift, only this one gives $|\chi_S| = 1$ for every state (100% of states at all 61 irreducible $\mathbf{Q}$ in GaN, against at most 35% for the others). The wrong pairing produced a fictitious ring of fractional parity around $\Gamma$.

**Selection rule.** The phonon perturbation transforms like its mode, $\sigma_h\,\partial V_\nu\,\sigma_h^{-1} = p_\nu\,\partial V_\nu$. Inserting $\sigma_h^\dagger\sigma_h = 1$ twice,

$$\mathcal{G}_{\beta\lambda\nu} = \langle\beta|\sigma_h^\dagger(\sigma_h\partial V_\nu\sigma_h^{-1})\sigma_h|\lambda\rangle = p^*_\beta\, p_\nu\, p_\lambda\, \mathcal{G}_{\beta\lambda\nu}, \tag{9}$$

so $\mathcal{G}_{\beta\lambda\nu} \neq 0$ only if $p_\beta p_\nu p_\lambda = +1$: an even phonon connects excitons of equal parity, and an odd phonon connects excitons of opposite parity.

*Quantifying it* (`xphd selection-rule`). Each term $|\mathcal{G}_{\beta\lambda\nu}(\mathbf{q})|^2$ is classed as allowed, forbidden or undefined (a final state with $|\chi_\beta|$ below a tolerance). For the bright state of GaN at $\Gamma$, forbidden/allowed $= 5\times10^{-7}$. `where_forbidden.py` separates three fingerprints of any residual forbidden weight: a phonon swap (labels and couplings in different orders), a state mix (a near-degenerate opposite-parity partner, or $|\chi| < 0.95$), and a diffuse admixture $\varepsilon = (1-|\chi|)/2$, for which forbidden/allowed into a state scales as $\varepsilon$ state by state.

**Irreducible representations at $\mathbf{K}$ and the emission rule.** The little group at $\mathbf{K}$ is $C_{3h}$, which is Abelian and has six one-dimensional irreducible representations (Table 5). Time reversal maps $\mathbf{K}$ to $\mathbf{K}'$ and adds no degeneracy at $\mathbf{K}$. Which of A, $E_1$, $E_2$ labels a state depends on the $C_3$ axis (through Ga, N or the hexagon centre), because a Bloch sum at $\mathbf{K}$ acquires a phase from sites off the axis; the $\sigma_h$ parity does not. `phonon_irreps_K.py` and `irreps_check.py` let you choose the axis.

Table 5: Character table of $C_{3h}$, with $\omega = e^{2\pi i/3}$.

| | $E$ | $C_3$ | $C_3^2$ | $\sigma_h$ | $S_3$ | $S_3^5$ |
|---|---|---|---|---|---|---|
| A′ | 1 | 1 | 1 | 1 | 1 | 1 |
| A″ | 1 | 1 | 1 | $-1$ | $-1$ | $-1$ |
| $E_1'$ | 1 | $\omega$ | $\omega^2$ | 1 | $\omega$ | $\omega^2$ |
| $E_2'$ | 1 | $\omega^2$ | $\omega$ | 1 | $\omega^2$ | $\omega$ |
| $E_1''$ | 1 | $\omega$ | $\omega^2$ | $-1$ | $-\omega$ | $-\omega^2$ |
| $E_2''$ | 1 | $\omega^2$ | $\omega$ | $-1$ | $-\omega^2$ | $-\omega$ |

For the dark A″ exciton at $\mathbf{K}$ to emit, a phonon must carry away the momentum $\mathbf{K}$ while a photon is emitted. The exciton must be contained in the product of the phonon and the photon,

$$\Gamma_{\mathrm{exc}} \subset \Gamma_\nu \otimes \Gamma_{\mathrm{photon}}, \tag{10}$$

a form that is invariant under a change of the $C_3$ axis (the conjugated form renames the helicities but selects the same branch). In-plane light is E′ in $D_{3h}$ and splits at $\mathbf{K}$ into $E_1'\oplus E_2'$, one component per helicity. For A″, a photon of $E_1'$ symmetry requires an $E_2''$ phonon, and one of $E_2'$ symmetry requires an $E_1''$ phonon. Only $E_1''$ occurs among the six GaN

Table 6: GaN phonons at $\mathbf{K}$ and whether each can mediate the emission of the A″ exciton under Eq. (10).

| Branch | $\hbar\omega$ (meV) | $\omega$ (cm$^{-1}$) | Irrep | $\sigma_h$ | Mediates |
|---|---|---|---|---|---|
| 1 | 16.2 | 130.8 | $E''_1$ | odd | **yes** |
| 2 | 17.9 | 145.1 | A″ | odd | no |
| 3 | 28.7 | 231.4 | $E'_2$ | even | no |
| 4 | 42.4 | 341.8 | A′ | even | no |
| 5 | 93.7 | 756.3 | $E'_1$ | even | no |
| 6 | 100.4 | 810.3 | $E'_2$ | even | no |

branches at $\mathbf{K}$ (Table 6), so one branch mediates the emission and its light has a single helicity. Time reversal gives $\mathbf{K}'$ the opposite one.

*When is the sideband fully polarised?* Let the $C_3$ eigenvalues of the two flexural modes at $\mathbf{K}$ be $\phi_1, \phi_2$ (two of $1, \omega, \omega^2$) and that of the dark exciton $\varepsilon$. By Eq. (10) a photon of eigenvalue $\omega$ needs a phonon of eigenvalue $\varepsilon\omega^2$, and one of eigenvalue $\omega^2$ needs a phonon of $\varepsilon\omega$. Both helicities have a partner only if $\{\varepsilon\omega, \varepsilon\omega^2\} = \{\phi_1, \phi_2\}$, that is, only if $\varepsilon$ is the third eigenvalue. If $\varepsilon$ equals $\phi_1$ or $\phi_2$, one helicity is available and each valley's sideband is fully circularly polarised. Shifting the axis multiplies $\varepsilon, \phi_1, \phi_2$ by the same phase, so the criterion is axis-independent. In GaN the two flexural modes at $\mathbf{K}$ are sublattice-polarised (each moves one sublattice): $E''_1$ at 16.2 meV and A″ at 17.9 meV. The A″ exciton ($\varepsilon = 1$) coincides with the A″ mode, so only $E''_1$ mediates.

**Two classes of planar $D_{3h}$ monolayers.** In a planar honeycomb binary, each band edge is $\pi$-like ($p_z$, odd) or $\sigma$-like ($s$, $p_{x,y}$, even), and for an exciton dominated by one transition $\chi = p_c p_v$. The valence maximum is the anion $p_z$ state at $\mathbf{K}$ (odd), so the class is set by the conduction minimum (Table 7). *Class I*: both edges are $\pi$-like at $\mathbf{K}$; the lowest exciton is the bright, even E′ state at $\Gamma$ and its sidebands are in-plane optical replicas (E′ $\subset$E′$\otimes$E′; the flexural branch is forbidden). *Class II*: a $\sigma$-like conduction band at $\Gamma$ lies lowest; the lowest exciton is momentum-dark and odd at $\mathbf{K}$, on a band disconnected from the bright one, and its sideband needs an odd phonon at $\mathbf{K}$.

Table 7: The two classes of planar monolayers, with the two materials treated here.

| | VBM | CBM | lowest exciton | sideband phonon | valley-locked |
|---|---|---|---|---|---|
| hBN (I) | $\mathbf{K}$, N $p_z$, odd | $\mathbf{K}$, B $p_z$, odd | $\Gamma$, even, E′ | TO/LO, even | no |
| GaN (II) | $\mathbf{K}$, N $p_z$, odd | $\Gamma$, Ga $s$, even | $\mathbf{K}$, odd, A″ (3.429 eV) | $E''_1$, odd, 16.2 meV | yes, $\lvert P\rvert = 1.000$ |

## 10.4 Near-degenerate excitonic spaces

**Why near-degenerate states mix.** Let two states of opposite parity be separated by $\Delta$, and let the computed BSE Hamiltonian break $\sigma_h$ by a small $\delta$ between them (numerical noise). First-order mixing gives an admixture amplitude $\delta/\Delta$, that is, $|\chi| = 1 - 2(\delta/\Delta)^2$. For $\Delta = 0.13$ meV, $\delta \sim 10$ $\mu$eV already gives $|\chi| \approx 0.99$. Per-state parity then either counts a forbidden residue or excludes the states.

**Splitting by the excitonic space's parity eigenstates.** Group the states at each $\mathbf{Q}$ into excitonic spaces of energies closer than a tolerance (`xphd parity -manifold-tol`, 5 meV is a good start) and compute, with Eq. (7), the matrix $M_{mn} = \langle m|\sigma_h|n\rangle$ within each. For a closed excitonic space $M$ is unitary and Hermitian with eigenvalues $\pm 1$; its eigenvectors $v_i$ are the parity eigenstates the solver could equally have returned. The state $\partial V_\nu|\lambda\rangle$ projected on the excitonic space is $|\psi\rangle = \sum_n G_n|n\rangle$ with $G_n = \langle n|\partial V_\nu|\lambda\rangle$, the archive's amplitudes. Its weight on eigenvector $i$ is

$$W_i = |\langle v_i|\psi\rangle|^2 = \left|\sum_n (v_i)_n^* G_n\right|^2, \qquad \sum_i W_i = \sum_n |G_n|^2, \tag{11}$$

and $W_i$ is allowed or forbidden according to the eigen-parity $\mu_i = \pm 1$. Nothing is excluded for being mixed; only eigenvectors with $|\mu_i|$ below the tolerance remain undefined. The weights in each parity sector, $\langle\psi|\frac{1}{2}(1 \pm M)|\psi\rangle$, do not depend on the basis within the excitonic space. This is what `xphd selection-rule -rotate` evaluates.

*Rotated states.* An archive's final state at $\mathbf{Q}+\mathbf{q}$ is the image of its irreducible parent, Eq. (4). For spatial $R$, $R^{-1}\sigma_h R = \sigma_h$ on excitons, so the excitonic space matrix is the parent's; for an operation containing time reversal (antiunitary) it is $M^*$. The time-reversal flag per full-zone point is the one `xphd generate` uses: symmetry index $\geq n_{\mathrm{sym}}/(1 + t_{\mathrm{rev}})$.

**Closure diagnostics.** An eigen-parity short of $\pm 1$ means the excitonic space is not closed, so its partner is missing. There are three possible reasons, and a script for each.

- *Cut by the state count* (`manifold_closure.py`): compare the eigen-parities of each excitonic space cut at the archive's state count with those of the complete excitonic space from a parity run with more states. If they differ, generate the archives with more states (`-nexc`).
- *Partner beyond the tolerance*: widen `-manifold-tol` and see whether the eigen-parities move.
- *Window not closed* (`window_closure.py`): $\sigma_h$ maps a band partly outside the BSE window when its mirror partner lies outside it. Then the row norm $\sum_{j\in\mathrm{window}} |D_{ij}(\sigma_h;\mathbf{k})|^2 < 1$, and every exciton built on that band maps partly outside the BSE space.

## 10.5 Exciton–phonon linewidths

**Golden rule.** To first order in the coupling and in the dilute limit, the full width at half maximum of state $\lambda$ at $\mathbf{Q}$ is

$$\Gamma_{\lambda\mathbf{Q}}(T) = 2\pi \sum_{\beta\nu} \int_{\mathrm{BZ}} \frac{A_{\mathrm{uc}}\, d^2q}{(2\pi)^2} \, |\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q},\mathbf{q})|^2 \Big[ (N_{\nu\mathbf{q}} + 1)\, \delta\big(E_{\lambda\mathbf{Q}} - E_{\beta\mathbf{Q}+\mathbf{q}} - \hbar\omega_{\nu\mathbf{q}}\big) + N_{\nu\mathbf{q}}\, \delta\big(E_{\lambda\mathbf{Q}} - E_{\beta\mathbf{Q}+\mathbf{q}} + \hbar\omega_{\nu\mathbf{q}}\big) \Big], \tag{12}$$

the imaginary part of the Fan–Migdal self-energy with the final-state exciton occupation set to zero. Since the zone has area $(2\pi)^2/A_{\mathrm{uc}}$, the measure is $A_{\mathrm{uc}}\, d^2q/(2\pi)^2 = d^2\tilde{q}/\tilde{A}_{\mathrm{BZ}} = (2/\sqrt{3})\, d^2\tilde{q}$ for $\tilde{q}$ in units of $|\mathbf{b}|$, and $(1/N_q)\sum_{\mathbf{q}}$ on a mesh. States within 2 meV are treated as one excitonic space and their widths averaged; the 2.8 meV exchange-split bright doublet of GaN is reported state by state.

**Linear-triangle integration.** On the $24 \times 24$ mesh a Lorentzian or Gaussian $\delta$ would need a width large enough to make the result depend on it, and in two dimensions it blurs the kinematic thresholds. Instead, the $\delta$ functions are integrated analytically [12, 13]. Each mesh cell is split into two triangles, and in each the energy difference $f = E_\lambda - E_\beta \mp \hbar\omega$ is linear between its vertex values $f_1 \le f_2 \le f_3$. The values of a linear function over a triangle of area $A_T$ are distributed with density

$$\rho_T(\varepsilon) = \frac{1}{A_T} \int_T \delta(\varepsilon - f)\, d^2r = \begin{cases} \dfrac{2(\varepsilon - f_1)}{(f_2 - f_1)(f_3 - f_1)}, & f_1 \le \varepsilon < f_2, \\ \dfrac{2(f_3 - \varepsilon)}{(f_3 - f_2)(f_3 - f_1)}, & f_2 \le \varepsilon < f_3, \\ 0, & \text{otherwise}, \end{cases} \tag{13}$$

normalised to $\int \rho_T\, d\varepsilon = 1$. The integral of Eq. (12) is then $\sum_T w_T A_T \rho_T(0)$, with $w_T$ the (piecewise-constant) weight $|\mathcal{G}|^2(N+1)$ or $|\mathcal{G}|^2 N$ of that triangle. This is done on a mesh refined 15-fold, to $360 \times 360$, which needs the phonon and exciton energies there.

**Phonons on the refined mesh.** We diagonalise the DFPT dynamical matrices directly with `matdyn.x`, including the two-dimensional non-analytic correction that removes the spurious LO–TO splitting at $\Gamma$ [14]. They reproduce the source mesh to 1.7 $\mu$eV on all branches, apart from a 0.237 meV acoustic residue at $\Gamma$ that the acoustic sum rule removes.

**Exciton energies on the refined mesh.** Exciton bands are indexed by ascending energy, so where two bands cross, an indexed band has a kink. A global Fourier interpolant [15, 16] rings from every kink across the zone; at the $\mathbf{K}$ minimum of GaN it made the band 34% too steep and gave it a spurious twofold anisotropy. We therefore take each refined energy from a quadratic about the nearest BSE point,

$$E(u,v) = E_0 + g_1 u + g_2 v + A u^2 + B u v + D v^2, \tag{14}$$

with $E_0$ the BSE energy and $(u,v)$ the reduced offset. The six neighbours lie at $\pm(h,0)$, $\pm(0,h)$, $\pm(h,-h)$. Sums over opposite pairs cancel the odd terms and fix the curvature exactly:

$$S_1 \equiv E(h,0) + E(-h,0) - 2E_0 = 2Ah^2, \quad S_2 = 2Dh^2, \quad S_3 = 2(A - B + D)h^2, \tag{15}$$

so $A = S_1/2h^2$, $D = S_2/2h^2$, $B = (S_1 + S_2 - S_3)/2h^2$. Differences give three estimates of the gradient, $d_1 = E(h,0) - E(-h,0) = 2hg_1$, $d_2 = 2hg_2$ and $d_3 = E(h,-h) - E(-h,h) = 2h(g_1 - g_2)$, each up to third-order terms. Minimising $(d_1 - x)^2 + (d_2 - y)^2 + (d_3 - x + y)^2$ over $x = 2hg_1$, $y = 2hg_2$ gives

$$g_1 = \frac{2d_1 + d_2 + d_3}{6h}, \qquad g_2 = \frac{d_1 + 2d_2 - d_3}{6h}. \tag{16}$$

The residual is the odd, trigonal-warping part of the band, third order in the offset. The quadratic reproduces $E_0$ exactly and its neighbours to third order, and because it uses only nearest neighbours, a crossing affects only the cells next to it. Quadratic interpolation onto a refined mesh followed by linear-triangle integration is the hybrid tetrahedron scheme of MacDonald *et al.* [17, 18], here in two dimensions [19]. We centre each quadratic on a mesh point rather than fitting one per triangle through its vertices and edge midpoints, because such per-triangle quadratics are continuous but kinked at the vertices, and a band minimum always lies on a vertex.

*Tests.* On model bands with closed-form linewidths, the per-triangle scheme erred by up to 17% at a minimum and the present one by at most 2.4%. A crossing 150 meV above the minimum shifted the Fourier result by 35% and the present one by 2%. At the GaN $\mathbf{K}$ minimum, fitting the curvature to the first, second and third shells of BSE points gives 12.44, 12.26 and 12.18 eV $|\mathbf{b}|^{-2}$. `compare_band_interp.py` predicts every BSE point from the rest, for both schemes, so you can run the same comparison on your own data. A Wannier representation of the exciton [20] would treat crossings rigorously.

**Couplings.** $|\mathcal{G}|^2$ is constant within each coarse cell, except for the three acoustic branches in the cell around $\mathbf{q} = 0$, where the coupling vanishes at the cell's own mesh point. There we use the long-wavelength form $|\mathcal{G}|^2\, 2\hbar\omega = Cq^2$ (that is, $|\mathcal{G}|^2 \propto q$ for a linear branch), with $C$ fitted for each pair of initial and final states to the innermost shell of the source mesh. Phonons below 0.5 meV are excluded. In `xphd linewidth` this is `-acoustic-model q2 -n-shell 1`.

**Symmetry.** Each state's width is invariant under the point group, so Eq. (12) is evaluated at the irreducible $\mathbf{Q}$ (61 for $24 \times 24$, $\mathrm{D}_{3h}$ with time reversal) and unfolded.

**The band minimum in closed form.** At the global minimum (GaN: $\mathbf{K}$) no lower state exists, and the width comes only from absorption of acoustic phonons within the valley, into final states inside the first mesh cell, where band and coupling are extrapolated. Model the valley as $E(q) = E_{\min} + bq^2$, the branch as $\hbar\omega = \hbar c q$ and the coupling as $|\mathcal{G}|^2\, 2\hbar\omega = Cq^2$, with $q$ in units of $|\mathbf{b}|$, following the deformation-potential treatment of exciton–acoustic-phonon scattering [21–23]. Absorption from the bottom conserves energy when $E_{\min} + \hbar c q = E_{\min} + bq^2$, that is, at

$$q^* = \frac{\hbar c}{b}, \qquad \hbar\omega^* = \hbar c q^* = \frac{(\hbar c)^2}{b} \quad \big(= 2Mc^2 \text{ for } b = \hbar^2/2M\big). \tag{17}$$

Then, with the measure above and $d^2q = 2\pi q\, dq$,

$$\begin{aligned}\Gamma &= 2\pi \cdot \frac{2}{\sqrt{3}} \int 2\pi q\, dq\, \frac{Cq}{2\hbar c}\, N(\hbar c q)\, \delta\big(bq^2 - \hbar c q\big) = \frac{8\pi^2}{\sqrt{3}}\, \frac{Cq^{*2}}{2\hbar c}\, \frac{N(\hbar\omega^*)}{|2bq^* - \hbar c|} \\ &= \frac{4\pi^2}{\sqrt{3}}\, \frac{C\, q^{*2}}{(\hbar c)^2}\, N(\hbar\omega^*) = \frac{4\pi^2}{\sqrt{3}}\, \frac{C\, N(\hbar\omega^*)}{b^2},\end{aligned} \tag{18}$$

using $|2bq^* - \hbar c| = \hbar c$. Above $\hbar\omega^*/k_B$, $N \approx k_B T/\hbar\omega^*$ and $\Gamma \approx (4\pi^2/\sqrt{3})\, C k_B T/[b(\hbar c)^2]$ grows linearly in $T$; below it the width freezes out. For GaN, $b = 12.44$ eV $|\mathbf{b}|^{-2}$ and $\hbar\omega^* = 1.49$ meV ($q^* = 0.011\,|\mathbf{b}|$), with $C$ taken from the minimum-to-minimum coupling (TA couples at $10^{-3}$ of LA, and ZA not at all, by parity). Equation (18) gives 0.09, 10.6–11.4 and 45–48 meV at 5, 77 and 300 K, the range coming from extrapolating $C$ to $q \to 0$. The code gives 0.087, 10.4 and 50.5 meV, the excess at 300 K being about 6 meV of absorption beyond the first cell. At 77 K the width exceeds $\hbar\omega^*$, so the scattering is quasi-elastic and the line is not strictly Lorentzian.

*Leaving the state versus leaving the valley.* The absorbed phonon moves the exciton by $|\mathbf{q}^*| = 0.011\,|\mathbf{b}|$ inside its valley, with almost no change of energy. Excluding the first cell (`-acoustic-model none`) leaves the rate of scattering beyond it, which is the escape from the valley bottom: 0.044 meV at 77 K ($\hbar/\Gamma \approx 15$ ps).

**Off-diagonal self-energy and validity.** The off-diagonal elements of the exciton–phonon self-energy [24] change the lineshape through `xphd lineshape`; compare the largest off-diagonal element with the diagonal width of the state you care about before deciding whether they matter. For the highest states, widths reach several hundred meV (lifetimes $\sim 1$ fs, comparable to the band spacing), and there Eq. (12) no longer describes a sharp resonance.

## 10.6 Phonon-assisted luminescence and helicity

**Second-order emission.** An exciton $\beta$ at momentum $\mathbf{q}$ emits through a virtual bright state $\lambda$ at $\mathbf{Q} = 0$, emitting or absorbing a phonon $(\mu, \mathbf{q})$ that carries the momentum [25, 26]:

$$\begin{aligned}I^{\mathrm{PL}}(\omega; T) \propto \frac{1}{N_q} \sum_{\beta,\mathbf{q},\mu,\pm} \Bigg| \sum_{\lambda} \frac{T_\lambda\, \mathcal{G}_{\lambda\beta\mu}(\mathbf{q}, -\mathbf{q})}{E_{\beta\mathbf{q}} \mp \hbar\omega_{\mu\mathbf{q}} - E_{\lambda\mathbf{0}}} \Bigg|^2 \\ \times\, N_{\beta\mathbf{q}}(T_{\mathrm{exc}})\, F^{\pm}_{\mu\mathbf{q}}(T)\, \delta\big(\hbar\omega - E_{\beta\mathbf{q}} \pm \hbar\omega_{\mu\mathbf{q}}\big),\end{aligned} \tag{19}$$

with $T_\lambda$ the dipole of the bright state, $N_{\beta\mathbf{q}} \propto e^{-E_{\beta\mathbf{q}}/k_B T_{\mathrm{exc}}}$ and $F^{\pm} = N_{\mu\mathbf{q}} + \frac{1}{2} \pm \frac{1}{2}$. In GaN ($T = T_{\mathrm{exc}} = 10$ K, 15 states, six branches, 5 meV broadening) the spectrum below the direct line is a single satellite at 3.4129 eV, 16.1 meV below the dark $\mathbf{K}$ exciton (3.4290 eV), matching the $\mathrm{E}_1''$ branch at $\mathbf{K}$ (16.2 meV). Branch by branch, it exceeds every other branch by seven orders of magnitude.

**Helicity and valley.** Replace $T_\lambda$ by $\hat{\boldsymbol{\epsilon}}^*_\pm \cdot \mathbf{d}_\lambda$ with $\hat{\boldsymbol{\epsilon}}_\pm = (\hat{\mathbf{x}} \pm i\hat{\mathbf{y}})/\sqrt{2}$ and the complex Cartesian dipoles $\mathbf{d}_\lambda$ of the $\mathbf{Q} = 0$ excitons, and separate the emitters $\beta$ by valley. The degree of circular polarisation is $P = (I_+ - I_-)/(I_+ + I_-)$. Because $\hat{\boldsymbol{\epsilon}}_\pm$ form an orthonormal basis of the plane, for any amplitude vector $\mathbf{v}$,

$$|\hat{\boldsymbol{\epsilon}}^*_+ \cdot \mathbf{v}|^2 + |\hat{\boldsymbol{\epsilon}}^*_- \cdot \mathbf{v}|^2 = |v_x|^2 + |v_y|^2, \tag{20}$$

so $I_+ + I_-$ equals the helicity-summed spectrum exactly (a sum rule checked to $3.7 \times 10^{-16}$ in GaN). The result is $P = -1.0000$ from $\mathbf{K}$ and $+1.0000$ from $\mathbf{K}'$, with equal valley intensities to 0.12% (ratio 0.9988). **The dipoles must come from a single BSE solution**: with dipoles and eigenvectors from different runs, or with the valleys summed, the helicity is lost. In hBN (class I) no valley-locked satellite is expected, which is why hBN serves as the negative control.

**Exciton dipoles and oscillator strengths.** `xphd dipoles` builds $\mathbf{d}_\lambda = \sum_{vc\mathbf{k}} A^\lambda_{vc\mathbf{k}}\, \mathbf{d}_{cv}(\mathbf{k})$ from the single-particle dipoles of `ndb.dipoles` in the BSE window, rotated to the full zone like the envelopes (complex conjugated for time-reversed points). It is validated against yambo's oscillator strengths excitonic space by excitonic space: the ratio is the same constant (the unit conversion) for every excitonic space, with exchange-split pairs compared as sums. For the transport, `add_osc.py` writes the in-plane strengths $|d_x|^2 + |d_y|^2$ into the $\Gamma$ archive (field `osc_Q0`); $\sigma_h$-odd, $z$-polarised excitons receive none.

## 10.7 Real-time exciton Boltzmann transport

**Equation and rates.** For the occupations $F_i(t)$ with $i \equiv (\lambda, \mathbf{Q})$ [7, 27],

$$\frac{dF_i}{dt} = -\sum_j \Big[ P_{ij} F_i (1 + F_j) - P_{ji} F_j (1 + F_i) \Big], \tag{21}$$

$$P_{ij} = \frac{2\pi}{\hbar} \frac{1}{N_q} \sum_\nu |\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}, \mathbf{q})|^2 \Big[ (N_{\nu\mathbf{q}} + 1)\, \delta(E_i - E_j - \hbar\omega_{\nu\mathbf{q}}) + N_{\nu\mathbf{q}}\, \delta(E_i - E_j + \hbar\omega_{\nu\mathbf{q}}) \Big], \tag{22}$$

with $\mathbf{q} = \mathbf{Q}_j - \mathbf{Q}_i$. With the matrix elements, phonons and triangle weights of Sec. 10.5, the row sums are the linewidths, $\sum_j P_{ij} = \Gamma_i/\hbar$. They agree to better than 0.1% with Eq. (12) evaluated on the same mesh (without the in-cell acoustic channel, which moves an exciton within its own cell and cannot move population between mesh states). The rate matrix couples different momenta, so it is assembled on the full zone: $576 \times 5 = 2880$ states for GaN.

**Detailed balance.** A transition $i \to j$ by emission is integrated over the triangle containing $\mathbf{q}$, and the reverse $j \to i$ by absorption over the one containing $-\mathbf{q}$. The two energy-conserving surfaces are identical by time reversal, but linear interpolation gives them different areas $A_{ij}^{\mathrm{em}} \neq A_{ji}^{\mathrm{abs}}$, and the system then accumulates particles and never equilibrates. With the symmetrised weight $B_{ij} = \frac{1}{2}(A_{ij}^{\mathrm{em}} + A_{ji}^{\mathrm{abs}})$,

$$P_{ij} = B_{ij}|\mathcal{G}|^2(N+1), \quad P_{ji} = B_{ij}|\mathcal{G}|^2 N \;\Rightarrow\; \frac{P_{ij}}{P_{ji}} = \frac{N+1}{N} = e^{\hbar\omega/k_BT} = e^{(E_i - E_j)/k_BT} \tag{23}$$

exactly, for every energy-conserving pair. The rates are written without dividing by $N$, which would amplify weakly populated channels. Particle number is conserved to $10^{-9}$.

*Stationary state.* With $F = 1/(e^{(E-\mu)/k_BT} - 1)$, one has $F/(1+F) = e^{-(E-\mu)/k_BT}$, so $P_{ij}F_i(1+F_j) = P_{ji}F_j(1+F_i)$ is exactly the ratio above. Every pair is balanced, and the Bose–Einstein distribution is stationary (verified to $10^{-16}$); at these densities it is Boltzmann.

**Refining the rates.** The full matrix cannot be refined like a linewidth (each refined $\mathbf{q}$ is a new state, and the size grows as the fourth power of the refinement), but its row sums are ordinary zone integrals. With $s_i$ the ratio of the row sum on a fivefold finer mesh to the coarse one, the matrix is rescaled as $P_{ij} \to P_{ij}\sqrt{s_i s_j}$, which leaves $P_{ij}/P_{ji}$ unchanged. A one-sided rescaling $P'_{ij} = s_i P_{ij}$ would not: detailed balance $P'_{ij}F_i = P'_{ji}F_j$ then gives $F_i/F_j = (s_j/s_i)\, e^{-(E_i-E_j)/k_BT}$, that is, $F_i \propto e^{-E_i/k_BT}/s_i$, which is the wrong equilibrium. The thermalisation time is set by the refined row sums, and the branching ratios vary smoothly with $\mathbf{q}$.

**Initial condition, integration, diagnostics.** Optical injection fills $\mathbf{Q} = 0$: $F_\lambda(0) \propto |\mu_\lambda|^2 \exp[-(E_\lambda - E_{\mathrm{pump}})^2/2\sigma^2]$, normalised to the injected density. The results are unchanged from $10^{-4}$ to $10^{-2}$ per state (linear regime). Equation (21) is stiff and is integrated with an implicit BDF solver (relative and absolute tolerances $10^{-7}$ and $10^{-12}$); occupations far above the minimum can dip to $-10^{-15}$, an integrator artefact. Two decompositions are reported: the filling rate $\sum_j P_{ji}^{(c)} F_j$ of a state by channel $c$, and the aggregate flux $\sum_{ij} F_i P_{ij}^{(c)}$. A channel can fill many sparsely populated states yet carry little flux.

**Results and the finite-temperature equilibrium.** GaN at 77 K: the population relaxes to 4.26 meV above the minimum, against a Boltzmann value of 4.09 meV on the same states. Both lie below $k_BT = 6.6$ meV, the value for a two-dimensional continuum, because the nearest mesh states around $\mathbf{K}$ lie $bh^2 = 22$ meV above it. Of the population, 35, 50, 90 and 99% reaches the $\mathbf{K}$, $\mathbf{K}'$ valleys (within $0.12\,|\mathbf{b}|$) by 30, 41, 98 and 172 fs. It then escapes the valley bottom at 0.044 meV (15 ps), although each $\mathbf{K}$ state loses coherence in 63 fs. The valleys fill mainly through the fourth branch in its in-plane, LA-like form: the valley and the feeding states are both $\sigma_h$-odd, so the phonon must be even.

At a temperature where $k_BT$ is comparable to the energy of the first shell of mesh points around the minimum, the equilibrium on the mesh is not concentrated on the minimum's own point. A light exciton at 300 K can hold only $\sim 20\%$ there; `check_bte.py` prints the share at the minimum's point and in each valley against Boltzmann on the same states. A run has settled when it matches Boltzmann, and equal emission and absorption flux for every branch is the same statement.

## 10.8 Validation summary

| Check | Result (GaN) | Where |
|---|---|---|
| Exciton parity, $\lvert\chi\rvert = 1$ | 100% of states | Sec. 10.3 |
| Selection rule, forbidden/allowed | $5 \times 10^{-7}$ | Sec. 10.3 |
| Star uniformity $T_{\max}/T_{\min}$ | 1.0001 | Sec. 10.2 |
| Excitonic-space-resolved symmetry, TR / $C_3$ | $1.3/3.3 \times 10^{-3}$ | Sec. 10.2 |
| Phonons, `matdyn` against the source mesh | 1.7 $\mu$eV | Sec. 10.5 |
| $\mathbf{K}$ minimum against the closed form, 77 K | 10.4 against 10.6–11.4 meV | Sec. 10.5 |
| Helicity sum rule; $P$ per valley | $3.7 \times 10^{-16}$; $\mp 1.0000$ | Sec. 10.6 |
| BTE row sums against linewidths | $< 0.1\%$ | Sec. 10.7 |
| BTE detailed balance; stationarity | machine precision; $10^{-16}$ | Sec. 10.7 |
| BTE end state against Boltzmann, 77 K | 4.26 against 4.09 meV | Sec. 10.7 |

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